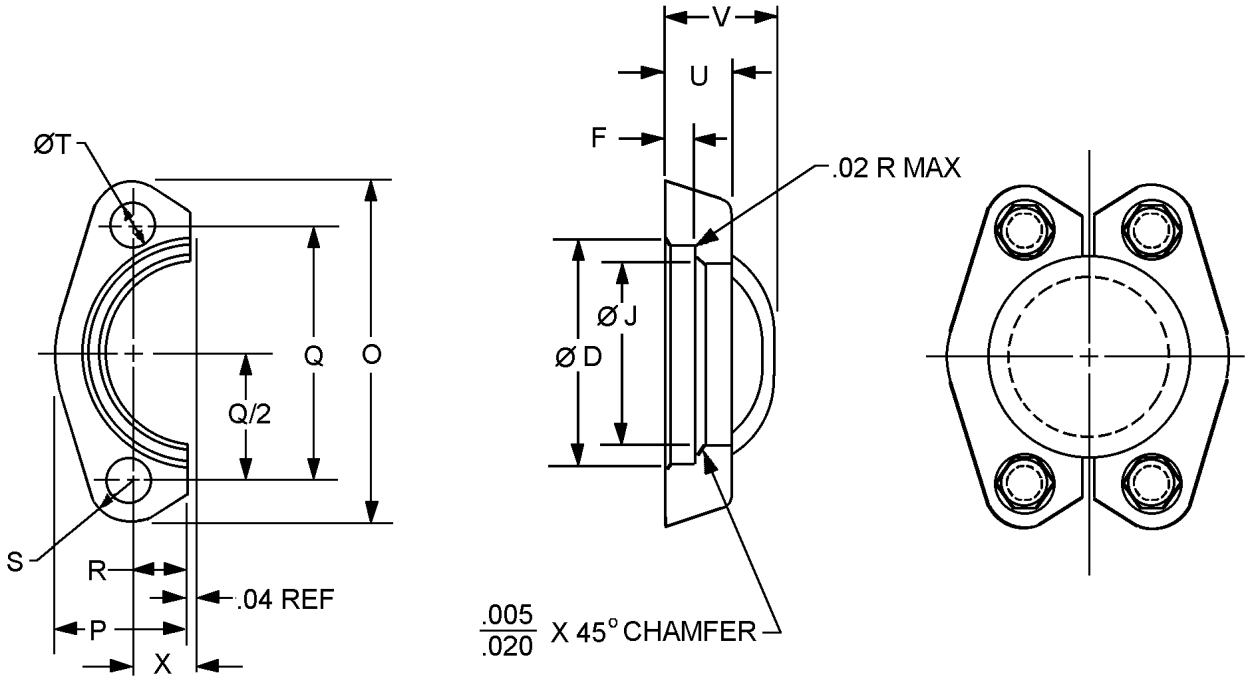
INCH-POUND
MIL-DTL-52525/16D
20 November 2008
SUPERSEDING
MIL-DTL-52525/16C
30 June 1998

DETAIL SPECIFICATION SHEET

FITTINGS, CLAMP-HALVES, FOR 4-BOLT SPLIT-FLANGE, SAE-J518, TYPE SPLIT-FLANGE CLAMP (SFC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-52525.



Inches	mm
.005	0.13
.02	0.5
.020	0.51
.04	1.0

FIGURE 1. Clamp-half hydraulic 4-bolt split-flange type.

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Part or Identifying Number (PIN)	Flange size inches (mm)	D (see note 4) inches (mm) ±.010 (0.25)	F inches (mm) ±.005 (0.13)	J inches (mm) ±.010 (0.25)
M52525/16-8	.500 (12.70)	1.219 (30.96)	.245 (6.22)	.955 (24.26)
M52525/16-12	.750 (19.05)	1.531 (38.89)	.245 (6.22)	1.265 (32.13)
M52525/16-16	1.000 (25.40)	1.781 (45.24)	.295 (7.49)	1.515 (38.48)
M52525/16-20	1.250 (31.75)	2.031 (51.59)	.295 (7.49)	1.720 (43.69)
M52525/16-24	1.500 (38.10)	2.406 (61.11)	.295 (7.49)	2.000 (50.80)
M52525/16-32	2.000 (50.80)	2.844 (72.24)	.355 (9.02)	2.470 (62.74)
M52525/16-40	2.500 (63.50)	3.344 (84.94)	.355 (9.02)	2.950 (74.93)
M52525/16-48	3.000 (76.20)	4.031 (102.39)	.355 (9.02)	3.580 (90.93)
M52525/16-56	3.500 (88.90)	4.531 (115.09)	.422 (10.72)	4.030 (102.36)
M52525/16-64	4.000 (101.60)	5.031 (127.79)	.422 (10.72)	4.530 (115.06)

PIN	O inches (mm)		P inches (mm) ±.030 (0.76)	Q inches (mm) ±.010 (0.25)	R inches (mm)
	max	min			
M52525/16-8	2.16 (54.86)	2.09 (53.09)	.86 (21.84)	1.500 (38.10)	.31 (7.87)
M52525/16-12	2.59 (65.79)	2.53 (64.26)	.98 (24.89)	1.875 (47.63)	.40 (10.16)
M52525/16-16	2.78 (70.61)	2.72 (69.09)	1.11 (28.19)	2.062 (52.37)	.48 (12.19)
M52525/16-20	3.16 (80.26)	3.09 (78.49)	1.39 (35.31)	2.312 (58.72)	.56 (14.22)
M52525/16-24	3.72 (94.49)	3.66 (92.96)	1.58 (40.13)	2.750 (69.85)	.67 (17.02)
M52525/16-32	4.06 (103.12)	3.94 (100.08)	1.86 (47.24)	3.062 (77.77)	.81 (20.57)
M52525/16-40	4.56 (115.82)	4.44 (112.78)	2.09 (53.09)	3.500 (88.90)	.96 (24.38)
M52525/16-48	5.38 (136.65)	5.25 (133.35)	2.53 (64.26)	4.188 (106.38)	1.18 (29.97)
M52525/16-56	6.06 (153.92)	5.94 (150.88)	2.70 (68.58)	4.750 (120.65)	1.34 (34.04)
M52525/16-64	6.44 (163.58)	6.31 (160.27)	2.95 (74.93)	5.125 (130.18)	1.49 (37.85)

PIN	S inches (mm)	T inches (mm) ±.010 (0.25)	U inches (mm)	V inches (mm)	X inches (mm) ±.010 (0.25)
M52525/16-8	.31 (7.87)	.344 (8.74)	.50 (12.70)	.75 (19.05)	.344 (8.74)
M52525/16-12	.34 (8.64)	.406 (10.31)	.56 (14.22)	.88 (22.35)	.438 (11.13)
M52525/16-16	.34 (8.64)	.406 (10.31)	.62 (15.75)	.94 (23.88)	.515 (13.08)
M52525/16-20	.41 (10.41)	.469 (11.91)	.56 (14.22)	.88 (22.35)	.594 (15.09)
M52525/16-24	.47 (11.94)	.531 (13.49)	.62 (15.75)	1.00 (25.40)	.703 (17.86)
M52525/16-32	.47 (11.94)	.531 (13.49)	.62 (15.75)	1.03 (26.16)	.844 (21.44)
M52525/16-40	.50 (12.70)	.531 (13.49)	.75 (19.05)	1.50 (38.10)	1.000 (25.40)
M52525/16-48	.56 (14.22)	.656 (16.66)	.88 (22.35)	1.62 (41.15)	1.219 (30.96)
M52525/16-56	.62 (15.75)	.656 (16.66)	.88 (22.35)	1.12 (28.45)	1.375 (34.93)
M52525/16-64	.62 (15.75)	.656 (16.66)	1.00 (25.40)	1.38 (35.05)	1.531 (38.89)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. D and J to be concentric within .010 inch (0.25 mm) of full indicator movement.
4. All dimensions not shown shall be in accordance with SAE-J518.

FIGURE 1. Clamp-half hydraulic 4-bolt split-flange type - Continued.

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REQUIREMENTS:

Fittings shall be as specified on figure 1 and in tables I and II.

The clamp-halves described herein are for use with MIL-DTL-52471/7, type 100R(X), double wire braid hose and clamp-halves, flange to port in accordance with MIL-DTL-52525/17.

If fittings are to be used on an oxygen hose see, MIL-DTL-52525.

Materials and finishes shall be in accordance with MIL-DTL-52525 and in table I. All finishes shall be capable of withstanding 96 hours minimum of salt spray.

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TABLE I. Material and finish identification codes. 1/ 2/

PIN code material/plating finish	Material	Plating finish
A	Aluminum alloy 6061 or 7075	Anodize in accordance with MIL-A-8625, type II.
AN		Anodize in accordance with MIL-A-8625, type II and NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type 2, class A.
BN		Bare aluminum with NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
C	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2.
CN		Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
YC	Chrome-molybdenum steel alloy 4130	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2.
YN		Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
F	Steel	NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
FN	Chrome-molybdenum steel alloy 4130	
G	Steel	Zinc plating with colorless passivate in accordance with ASTM B633, type V, Fe/Zn 25.
YG	Chrome-molybdenum steel alloy 4130	
H	Steel	Zinc phosphate finish in accordance MIL-DTL-16232, type Z, class1. 3/
YH	Chrome-molybdenum steel alloy 4130	
J	Steel	Zinc plating with chromate conversion in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5.
YJ	Chrome-molybdenum steel alloy 4130	
K	Nickel-copper alloy UNS N04400	No additional finish.
M	High-chromium nickel alloy UNS N06690	No additional finish.
N	Steel	Zinc aluminum in accordance with ASTM F1136, grade 3, NC.
YN	Chrome-molybdenum steel alloy 4130	
P	Steel	Zinc plating with colorless passivate in accordance with ASTM B633, type VI, Fe/Zn 5.
YP	Chrome-molybdenum steel alloy 4130	
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7.
SN		Passivation above and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
T	Titanium	Anodize in accordance with SAE-AMS2488, type 2.
V	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 1.
W	Steel	Any zinc plating above.
YW	Chrome-molybdenum steel alloy 4130	
Z	Steel	Any zinc plating above with NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
YZ	Chrome-molybdenum steel alloy 4130	

1/ All materials and finishes shall be in accordance with MIL-DTL-52525.
2/ Embitterment test need not be run.
3/ Zinc phosphate finish is hexavalent chromium free.

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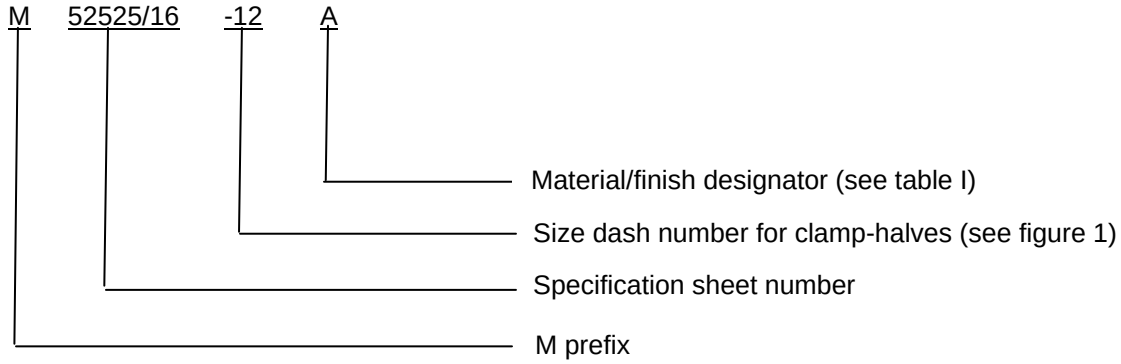
Maximum operating pressure shall be as specified in table II. Maximum operating pressures are for low carbon steel fittings consult manufacturer for values on other materials.

TABLE II. Maximum operating pressures of fittings. 1/

SAE dash size	PIN	Working pressure (max)		Burst pressure	
		psi	MPa	psi	MPa
-8	M52525/16-8	5000	34.5	10000	69
-12	M52525/16-12	5000	34.5	10000	69
-16	M52525/16-16	5000	34.5	10000	69
-20	M52525/16-20	4000	27.6	10000	69
-24	M52525/16-24	3000	20.7	10000	69
-32	M52525/16-32	3000	20.7	10000	69
-40	M52525/16-40	2500	17.3	7100	49
-48	M52525/16-48	2000	13.8	6000	41.4
-56	M52525/16-56	500	3.4	2000	13.8
-64	M52525/16-64	500	3.4	2000	13.8

1/ Metric equivalents are given for information only.

PIN: The PIN consists of the letter “M” the specification sheet number, a dash, a number for clamp size, and a material finish designator.



PIN example: M52525/16-12A, describes a 4-bolt split-flange .750 inch (19.05 mm), aluminum with an anodized finish.

Color identification. Color identification shall be in accordance with SAE-AS4841.

Marking shall include the manufacturer’s name or trademark, size, and hose identifier R10. Location of marking is optional.

Supersession data in table III is for reference only. SAE style fittings may not have the same material/finish as specified in MIL-DTL-52525.

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TABLE III. Supersession data. 1/

PIN	SAE PIN
M52525/16-8	SAE J518-1 8 1161
M52525/16-12	SAE J518-1 12 1161
M52525/16-16	SAE J518-1 16 1161
M52525/16-20	SAE J518-1 20 1161
M52525/16-24	SAE J518-1 24 1161
M52525/16-32	SAE J518-1 32 1161
M52525/16-40	SAE J518-1 40 1161
M52525/16-48	SAE J518-1 48 1161
M52525/16-56	SAE J518-1 56 1161
M52525/16-64	SAE J518-1 64 1161

1/ Material and finish designators are omitted from PIN's see table I, SAE-J518, and SAE-J846.

To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when the other materials and finishes specified in this document cannot meet performance requirements.

Legacy O-ring information. Table IV lists legacy information to help in determining the correct O-ring to use on 4-bolt split flange fittings and clamp halves.

TABLE IV. O-ring legacy table.

Dash	PIN's	Width (T or d2)	ID (d1)	Comments
-8	AN6227-15	.0934	.750	Original MIL-DTL-52525 call out which is cancelled to SAE-AS28775
-12	AN6227-19	.0934	1.000	Original MIL-DTL-52525 call out which is cancelled to SAE-AS28775
-16	AN6227-24	.125	1.313	Original MIL-DTL-52525 call out which is cancelled to SAE-AS28775
-20	AN6227-27	.125	1.500	Original MIL-DTL-52525 call out which is cancelled to SAE-AS28775
-24	AN6230-3	.125	1.875	Original MIL-DTL-52525 call out which is cancelled to SAE-AS28775
-32	AN6230-6	.125	2.250	Original MIL-DTL-52525 call out which is cancelled to SAE-AS28775
-8	SAE-J120-210	.139 +/-.004	.734	Cancelled without replacement should be cancelled to SAE-J515.
-12	SAE-J120-214	.139 +/-.004	.984	Cancelled without replacement should be cancelled to SAE-J515.
-16	SAE-J120-219	.139 +/-.004	1.296	Cancelled without replacement should be cancelled to SAE-J515.
-20	SAE-J120-222	.139 +/-.004	1.484	Cancelled without replacement should be cancelled to SAE-J515.
-24	SAE-J120-225	.139 +/-.004	1.859	Cancelled without replacement should be cancelled to SAE-J515.
-32	SAE-J120-228	.139 +/-.004	2.234	Cancelled without replacement should be cancelled to SAE-J515

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TABLE IV. O-ring legacy table - Continued.

Dash	PIN's	Width (T or d2)	ID (d1)	Comments
-8	SAE-J515-210	.1389 +/-.004	.734	Replacement for SAE-J120
-12	SAE-J515-214	.1389 +/-.004	.98	Replacement for SAE-J120
-16	SAE-J515-219	.1389 +/-.004	1.296	Replacement for SAE-J120
-20	SAE-J515-222	.1389 +/-.004	1.484	Replacement for SAE-J120
-24	SAE-J515-225	.1389 +/-.004	1.859	Replacement for SAE-J120
-32	SAE-J515-228	.1389 +/-.004	2.234	Replacement for SAE-J120
-8	SAE-AS28775-210	.135-.143	.728-.0740	Replacement for SAE-AS6227 and SAE-AS6230
-12	SAE-AS28775-214	.135-.143	.978-.0990	Replacement for SAE-AS6227 and SAE-AS6230
-16	SAE-AS28775-219	.135-.143	1.290-1.302	Replacement for SAE-AS6227 and SAE-AS6230
-20	SAE-AS28775-222	.135-.143	1.478-1.490	Replacement for SAE-AS6227 and SAE-AS6230
-24	SAE-AS28775-328	.205-.215	1.840-1.860	Replacement for SAE-AS6227 and SAE-AS6230
-32	SAE-AS28775-331	.205-.215	2.215-2.235	Replacement for SAE-AS6227 and SAE-AS6230

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-52525, this document references the following:

MIL-A-8625	ASTM B695	SAE-AMS-QQ-P-416
MIL-DTL-16232	ASTM F1136	SAE-AS4841
MIL-DTL-52471/7	SAE-AMS2417	SAE-AS28775
MIL-DTL-52525/17	SAE-AMS2488	SAE-J515
MIL-DTL-81706	SAE-AMS2700	SAE-J518
ASTM B633	SAE-AMS-C-81562	SAE-J846

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2007-084)

Review activities:
Army - CR4
Navy - AS, CG, MC, SA, YD

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